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STEP 2 - Similarity matrix between all occurrences of the word (last layer)
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mean similarity, SAME sense      : 0.773
mean similarity, DIFFERENT sense : 0.464
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# -----
banner("STEP 2 - Similarity matrix between all occurrences of the word (last layer)")
# -----
last = F.normalize(vecs[:, -1, :], dim=-1)
sim = (last @ last.T).numpy()
short = [f"{lab[:5]} {i+1}" for i, lab in enumerate(labels)]

fig, axes = plt.subplots(1, 2, figsize=(13, 5.5))
first = F.normalize(vecs[:, 0, :], dim=-1)
for ax, M, title in [(axes[0], (first @ first.T).numpy(), "Layer 0 (input embeddings)",
                        (axes[1], sim, f"Layer {n_layers} (last layer)"]):
    im = ax.imshow(M, cmap="viridis", vmin=min(sim.min(), 0.3), vmax=1)
    ax.set_xticks(range(len(short)), short, rotation=90)
    ax.set_yticks(range(len(short)), short)
    ax.set_title(title)
    plt.colorbar(im, ax=ax, fraction=0.046)
fig.suptitle(f"Cosine similarity between occurrences of '{TARGET}'\n"
            "Two blocks appear on the right: the model has discovered the two senses")
show("ex1_step2_similarity_matrix")

river = np.array(labels) == "river"
same = (sim[np.ix_(river, river)].sum() - river.sum() + sim[np.ix_(~river, ~river)].sum() - (~river).sum())
same /= (river.sum() * (river.sum() - 1) + (~river).sum() * ((~river).sum() - 1))
diff = sim[np.ix_(river, ~river)].mean()
print(f" mean similarity, SAME sense      : {same:.3f}")
print(f" mean similarity, DIFFERENT sense : {diff:.3f}")
```